Notes: Houston and James (JF, 1996)

Bank Information Monopolies and the Mix of Private and Public Debt Claims

Contents

1 Big picture

- **Research question.** How do large publicly traded firms choose between bank debt, other private debt, and public debt? More specifically, does the possibility that a bank acquires an information monopoly limit firms' reliance on bank borrowing?
- **Main idea.** Bank lending can be valuable because banks monitor, produce information, and renegotiate more flexibly than dispersed public bondholders. But an exclusive relationship with one bank can also create hold-up: once the bank learns private information about the borrower, switching lenders becomes costly and the bank may capture rents.
- **Empirical object.** The paper uses hand-collected debt-structure data from annual reports and 10-K filings for 250 exchange-traded firms in 1980, 1985, and 1990. The key dependent variable is the fraction of total debt supplied by banks.
- **Central prediction.** If monitoring benefits dominate, firms with greater growth opportunities and more information problems should use more bank debt. If bank hold-up is important, firms with growth opportunities should avoid relying on a single bank, unless they can mitigate hold-up by using multiple banks or public debt.
- **Main finding.** Among firms with a single bank relationship, reliance on bank debt is negatively related to growth opportunities. Among firms with multiple bank relationships, the relation becomes positive.
- **Public debt as a discipline device.** The paper also finds evidence that public debt can reduce bank bargaining power, though public debt and multiple-bank relationships are highly correlated in the sample.
- **Other private debt is not equivalent to bank debt.** Nonbank private debt does not appear to mitigate bank information monopoly problems in the same way as multiple bank relationships or public debt.
- **Economic takeaway.** Banks are useful monitors, but relationship lending has a shadow cost. Firms with valuable growth options avoid becoming locked into a single informed lender.

2 Incremental contribution of the paper

- **From bank-loan announcement effects to debt-mix choice.** Earlier empirical work showed that bank loan announcements can convey good news, suggesting banks are special. Houston and James study a different question: how much of a firm's debt should come from banks once the costs of bank control are considered?
- **Detailed debt-structure data.** The paper constructs a firm-level database that separates bank debt, public debt, commercial paper, and other private debt. This is important because standard databases generally do not identify debt sources in sufficient detail.
- **Large-firm focus.** The sample consists of large publicly traded firms that plausibly have access to public debt markets. This makes the private-versus-public debt choice especially meaningful.
- **Information monopoly test.** The paper does not simply ask whether banks are useful. It tests the Rajan-style prediction that a single bank relationship can become costly when the firm has many growth opportunities.

- **Multiple banks as an empirical margin.** The paper uses the number of banking relationships as a way to distinguish bank monitoring benefits from bank hold-up costs.
- **Comparison with public and nonbank private debt.** The analysis asks whether public debt and other private debt perform the same role as multiple banks in limiting lender bargaining power. The answer is: public debt may help, but other private debt does not appear to substitute for bank relationships.
- **Implication for relationship lending.** The findings make clear that the value of bank relationships depends on market structure. A bank relationship is not simply good or bad; its effect depends on whether the firm is locked into one lender.

3 Data and measurement

3.1 Sample construction

The paper begins with 250 randomly selected firms from the universe of NYSE, AMEX, and NASD firms with COMPUSTAT and CRSP data and long-term debt outstanding in fiscal year 1980. Financial institutions and regulated utilities are excluded. The authors collect debt-structure information for fiscal years 1980, 1985, and 1990.

- **Firm-years.** The maximum sample is 250 firms in 1980, 193 firms in 1985, and 161 firms in 1990.
- **Sources.** Financial data come from COMPUSTAT and CRSP; debt-structure information comes from annual reports and SEC 10-K filings.
- **Attrition.** Firms leave the sample because of acquisitions, distress-related delistings, going-private transactions, and other exits.
- **Why large firms?** Large traded firms are most likely to face a real choice between public and private debt. They are also the firms where any decline in traditional bank lending should be most visible.

3.2 Debt-structure variables

The key debt categories are measured from the notes to financial statements.

- **Bank debt.** Defined broadly as debt identified as bank borrowing plus private debt where the lender is not identified. This choice may overstate commercial bank debt, but it avoids incorrectly dropping bank loans that are not labeled explicitly.
- **Public debt.** Publicly issued long-term debt claims.
- **Commercial paper.** Short-term public market borrowing.
- **Other private debt.** Privately placed debt where the lender is identified as nonbank, such as insurance companies, finance companies, or pension funds.
- **Total debt denominator.** Short-term plus long-term debt, excluding mortgage debt, industrial revenue bonds, loans from stockholders, and capitalized leases.
- **Multiple banking relationships.** A firm is classified as having multiple bank relationships when the annual report or 10-K indicates borrowing from more than one bank. If the number of banks cannot be determined, the observation is assigned missing for that variable.

3.3 Firm characteristics

The empirical tests relate bank debt reliance to variables that proxy for information asymmetry, growth opportunities, credit quality, and access to public markets.

- **Growth opportunities.** Market-to-book ratio and R&D-to-sales ratio.
- **Firm size.** Log of total assets.
- **Leverage.** Short-term plus long-term debt divided by total assets.
- **Coverage.** Interest coverage ratio, used as a credit-quality proxy.
- **Public debt indicator.** Equals one when the firm has public debt outstanding.
- **Other private debt indicator.** Equals one when the firm has nonbank private debt outstanding.
- **Year dummies.** Indicators for 1985 and 1990 capture broad time patterns in the sample.

4 Theory and empirical specifications

4.1 Why bank debt can be valuable

Private lenders, and especially banks, can reduce adverse selection and moral hazard. Concentrated debt ownership creates incentives to monitor, avoids duplication of monitoring costs, and permits easier renegotiation than public debt. These benefits imply that firms with more information problems, more intangible assets, and more growth opportunities may demand more bank debt.

4.2 Why bank debt can be costly

The same information production that makes banks valuable can also create an information monopoly. Once a bank becomes informed about a borrower, the firm may find it difficult to switch lenders. A single bank can then extract rents during future renegotiation or refinancing. This can distort investment incentives, especially for growth firms whose future projects are valuable.

4.3 Predictions

The paper contrasts two views.

- **Monitoring-benefit view.** Bank debt should be increasing in market-to-book and R&D intensity because banks are useful when information problems and agency conflicts are severe.
- **Information-monopoly view.** For firms with a single bank relationship, bank debt may be decreasing in market-to-book and R&D intensity because growth firms fear hold-up.
- **Multiple-bank mitigation.** If multiple banks reduce hold-up, then the relation between growth opportunities and bank debt should be less negative, or positive, for firms with multiple banks.
- **Public-debt mitigation.** Public debt may also reduce bank bargaining power by giving firms an arm's-length funding source.
- **Nonbank private debt test.** If any nonbank private lender could mitigate hold-up, then other private debt should interact with growth opportunities like public debt. The paper finds little evidence for this.

4.4 Baseline regression

The main empirical specification explains the fraction of debt from banks:

$$\frac{\text{Bank debt}_{it}}{\text{Total debt}_{it}} = \alpha + \beta_1 \text{Growth}_{it} + \beta_2 (\text{Multiple banks}_{it} \times \text{Growth}_{it}) + \beta_3 \text{Multiple banks}_{it} + \Gamma' X_{it} + \delta_t + \varepsilon_{it}.$$

The vector X_{it} includes controls such as coverage, leverage, and log assets. The growth variable is either market-to-book or R&D-to-sales. The key coefficient is the interaction between the growth proxy and the multiple-bank dummy.

- A negative β_1 means that among single-bank firms, growth opportunities are associated with lower bank-debt reliance.
- A positive β_2 means that multiple bank relationships weaken or reverse that negative relation.
- The paper estimates both OLS and double-sided Tobit specifications because the bank-debt share is bounded between zero and one.

4.5 Public and other private debt specifications

The paper also estimates specifications that replace the multiple-bank interaction with interactions involving public debt and other private debt:

$$\frac{\text{Bank debt}_{it}}{\text{Total debt}_{it}} = \alpha + \beta_1 \text{Growth}_{it} + \beta_2 (\text{Public debt}_{it} \times \text{Growth}_{it}) + \beta_3 \text{Public debt}_{it} + \Gamma' X_{it} + \delta_t + \varepsilon_{it}.$$

The analogous specification with nonbank private debt asks whether private lenders other than banks serve the same information-monopoly mitigation role.

5 Identification and interpretation

5.1 What the paper can identify

The evidence is best interpreted as a cross-sectional relationship between debt structure, bank relationships, and firm characteristics. The paper identifies whether firms that are more exposed to potential bank hold-up rely less on bank debt, and whether this relation changes when firms use multiple banks or public debt.

5.2 Why growth opportunities matter

Growth opportunities are central because the Rajan-style hold-up problem is most damaging when future investment projects are valuable. A bank that can capture part of the surplus from future projects reduces the firm's incentive to invest. Therefore, the strongest hold-up predictions concern high market-to-book and high R&D firms.

5.3 Why multiple banks matter

Multiple banks may mitigate hold-up because no single bank controls all relationship-specific information. Competition among informed lenders can reduce the rents any one lender can extract. Multiple-bank borrowing can also indicate that information about the firm is easier to verify or that the firm is large enough to maintain outside funding options.

5.4 Why public debt matters

Public debt can reduce bank control by giving the firm arm's-length lenders outside the bank relationship. However, the paper cautions that public debt and multiple banking relationships are strongly correlated, which makes it hard to separately identify their effects.

5.5 Measurement concerns

- **Bank debt may be overstated.** Debt from unidentified private lenders is classified as bank debt.
- **Multiple-bank status is imperfectly observed.** Some firm-years are assigned missing when the reports do not reveal whether the firm has one bank or multiple banks.
- **Public-market access is endogenous.** Firms with public debt are larger, older, and more levered, so public debt may proxy for broader access to capital markets.
- **Growth proxies are imperfect.** Market-to-book and R&D-to-sales proxy for growth opportunities and information problems, but they also correlate with industry and firm quality.

5.6 Interpretation of the main result

The negative relation between growth opportunities and bank-debt reliance among single-bank firms is the core evidence for information-monopoly costs. The positive interaction with multiple bank relationships suggests that bank debt becomes more attractive when the firm can avoid being locked into one informed lender.

6 Tables and figures

6.1 Tables 1 and 2: Financial characteristics and debt structure

Read:

- Table 1 reports summary statistics for all available firm-years in 1980, 1985, and 1990.
- Bank debt is the dominant debt source: across all observations, the mean bank-debt-to-total-debt ratio is 0.64 and the median is 0.77.
- Public debt is present but less dominant on average: the mean public-debt-to-total-debt ratio is 0.15 and the median is zero.
- The percentage of firms with public debt rises from 35 percent in 1980 to 46 percent in 1990.
- Table 2 repeats the summary statistics for the 161 firms with data in all three sample years.
- The surviving-firm sample looks similar to the full sample, so the main time-series patterns are not driven only by sample attrition.

Economic message: Even large traded firms rely heavily on bank debt. The paper is therefore not about small private firms only; bank debt remains central in the debt structure of many publicly traded corporations.

Table I

Descriptive Statistics Concerning Financial Characteristics and Debt Structure

The sample consists of observations from 1980, 1985, and 1990 for 250 randomly selected exchange and National Association of Securities Dealers (NASD) traded firms with long term debt outstanding in 1980. Financial information is from COMPUSTAT, 10K filings, and annual reports. Leverage is measured as short plus long term debt divided by total assets. The market to book ratio is the book value of liabilities and preferred stock plus the market value of common stock divided by the book value of assets. Bank debt is from the firm's annual report or 10K filing. Total debt equals short plus long term debt less industrial revenue bonds, mortgages, and capitalized leases. Other private debt is privately placed bonds and loans specified in the financial statements as being from nonbank sources.

	Overall		1980		1985		1990	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median
Assets (millions)	1502	112	906	82	1430	115	2666	225.1
Leverage	0.39	0.35	0.36	0.35	0.39	0.33	0.45	0.40
Coverage ratio	11.21	3.34	10.71	3.94	14.96	3.4	6.91	2.32
Market/book	1.38	1.08	1.30	1.01	1.32	1.17	1.62	1.07
Bank debt/total debt	0.64	0.77	0.66	0.82	0.66	0.80	0.59	0.69
Public debt/total debt	0.15	0.00	0.13	0.00	0.16	0.00	0.17	0.00
Commercial paper/total debt	0.02	0.00	0.01	0.00	0.02	0.00	0.03	0.00
Other private/total debt	0.13	0.01	0.16	0.01	0.11	0.00	0.11	0.01
Unused lines of credit/total bank debt	2.66	0.48	2.39	0.44	2.40	0.60	3.60	0.46
Unused lines of credit/assets	0.13	0.07	0.13	0.05	0.12	0.09	0.13	0.08
Percent with multiple banking relationships	66		59		73		68	
Percent with secured bank debt	31		31		28		34	
Percent with public debt	39		35		42		46	
Percent with commercial paper	7.6		5.2		8		12	

(a) Table 1: Financial characteristics and debt structure.

Table II

Descriptive Statistics Concerning the Financial Characteristics and Debt Structure for Firms with Information Available in All Years (N = 161)

The sample consists of observations from 1980, 1985, and 1990 for 250 randomly selected exchange and National Association of Securities Dealers (NASD) traded firms with long term debt outstanding in 1980. Financial information is from COMPUSTAT, 10K filings, and annual reports. Leverage is measured as short plus long term debt divided by total assets. The market to book ratio is the book value of liabilities and preferred stock plus the market value of common stock divided by the book value of assets. Bank debt is from the firm's annual report or 10K filing. Total debt equals short plus long term debt less industrial revenue bonds, mortgages, and capitalized leases. Other private debt is privately placed bonds and loans specified in the financial statements as being from nonbank sources.

	Overall		1980		1985		1990	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median
Assets (millions)	1835	136	1163	87	1676	136	2666	225.1
Leverage	0.39	0.36	0.37	0.35	0.35	0.33	0.45	0.40
Coverage ratio	12.15	3.15	13.84	4.03	15.75	3.32	6.91	2.32
Market/book	1.46	1.11	1.41	1.06	1.36	1.19	1.62	1.07
Bank debt/total debt	0.63	0.72	0.63	0.73	0.66	0.77	0.59	0.69
Public debt/total debt	0.16	0.00	0.15	0.00	0.16	0.00	0.17	0.00
Commercial paper/total debt	0.02	0.00	0.01	0.00	0.02	0.00	0.03	0.00
Other private/total debt	0.13	0.00	0.17	0.00	0.11	0.00	0.11	0.01
Unused lines of credit/total bank debt	2.75	0.47	3.12	0.48	1.58	0.52	3.60	0.46
Unused lines of credit/assets	0.11	0.07	0.11	0.05	0.10	0.09	0.13	0.08
Percent with multiple banking relationships	67		63		70		68	
Percent with secured bank debt	33		36		28		34	
Percent with public debt	41		37		41		46	
Percent with commercial paper	9		6		9		12	

(b) Table 2: Surviving-firm financial characteristics.

6.2 Table 3: Types of bank borrowing

Read:

- Table 3 decomposes bank borrowing into borrowing on lines of credit, term borrowing, and short-term borrowing.
- For the full sample, short-term bank borrowing is 0.36 of total debt on average, term borrowing is 0.28, and borrowing on lines is 0.22.
- The overall bank-debt share remains high across the full period and among surviving firms.
- The table makes clear that bank debt is not a single homogeneous instrument. It includes short-term notes, lines of credit, and term loans.

Economic message: Bank debt is important through both liquidity-oriented short-term facilities and longer-term loans. This matters because bank control and monitoring can operate through several different contract types.

6.3 Table 4: Multiple banking relationships

Read:

- Table 4 compares firms with one bank to firms with multiple bank relationships.
- Single-bank firms are much smaller: mean assets are about 138 million dollars, compared with about 1.93 billion dollars for multiple-bank firms.
- Multiple-bank firms use more bank debt as a share of total debt and as a share of assets.
- Single-bank firms use more other private debt.
- Multiple-bank firms are older, larger, and more likely to have public debt.

Economic message: Multiple banking relationships are strongly associated with firm size and capital-market access. The empirical challenge is to distinguish the effect of lender competition from the fact that multiple-bank firms are different firms.

Table III

Proportion of Total Debt from Commercial Banks Grouped by Type of Bank Loan

The sample consists of observations from 1980, 1985, and 1990 for 250 randomly selected exchange and National Association of Securities Dealers (NASD) traded firms with long term debt outstanding in 1980. Financial information is from COMPUSTAT, 10K filings, and annual reports. Leverage is measured as short plus long term debt divided by total assets. Borrowing on lines of credit equals borrowing on bank supplied revolving credit agreements or lines of credit. Term borrowing equals term loans from commercial banks and term loans not specified as being from nonbank lenders. Short-term borrowing (under 5 years) equals short-term loans from commercial banks and short-term loans not specified as being from nonbank lenders. Short-term loans include loans made as part of a line of credit.

	Entire Period		1980		1985		1990	
All Firms	Mean	Median	Mean	Median	Mean	Median	Mean	Median
Borrowing on lines/total debt	0.22	0.00	0.19	0.00	0.24	0.00	0.23	0.00
Term borrowing/total debt	0.28	0.09	0.28	0.05	0.30	0.13	0.25	0.00
Short-term borrowing/total debt	0.36	0.24	0.38	0.26	0.36	0.24	0.34	0.14
All bank borrowing/total debt	0.64	0.77	0.66	0.82	0.66	0.80	0.59	0.69
Surviving firms only (N = 161)								
Borrowing on lines/total debt	0.22	0.00	0.18	0.00	0.25	0.00	0.23	0.00
Term borrowing/total debt	0.27	0.09	0.27	0.10	0.29	0.10	0.25	0.00
Short-term borrowing/total debt	0.36	0.19	0.36	0.20	0.37	0.29	0.34	0.14
All bank borrowing/total debt	0.63	0.72	0.63	0.73	0.66	0.77	0.59	0.69

(a) Table 3: Bank borrowing by loan type.

Table IV

Descriptive Statistics Concerning the Financial Characteristics and Debt Structure for Firms Sorted by Whether they have Multiple Banking Relationships

The sample consists of observations from 1980, 1985, and 1990 for 250 randomly selected exchange and National Association of Securities Dealers (NASD) traded firms with long term debt outstanding in 1980. Financial information is from COMPUSTAT, 10K filings, and annual reports. Leverage is measured as short plus long term debt divided by total assets. The market to book ratio is the book value of liabilities and preferred stock plus the market value of common stock divided by the book value of assets. Bank debt is from the firm's annual report or 10K filing. Total debt equals short plus long term debt less industrial revenue bonds, mortgages, and capitalized leases. Other private debt is privately placed bonds and loans specified in the financial statements as being from nonbank sources. Investment equals capital expenditures as reported in COMPUSTAT. Fixed assets are plant, property, and equipment.

	Firms With One Bank		Firms with Multiple Banks	
	Mean	Median	Mean	Median
Assets (millions)	137.96*	26.78*	1933.64	251.38
Age (years since incorporated)	21*	24*	49	48
Leverage	0.36	0.34	0.40	0.37
Coverage ratio	12.44*	3.07	6.81	3.24
R&D/sales	0.041	0.017	0.032	0.018
Market/book	1.57	1.04	1.32	1.07
Bank debt/total debt	0.55*	0.75	0.69	0.82
Bank debt/assets	0.07*	0.13*	0.13	0.17
Public debt/total debt	0.13	0.00	0.18	0.00
Other private/total debt	0.23*	0.014	0.087	0.01
Investment/assets	0.07	0.05	0.08	0.07
Growth in assets over five years**	0.353*	0.218*	0.939	0.463
Return on assets (ROA)	0.044	0.095	0.092	0.095
Fixed assets/assets	0.262*	0.293*	0.35	0.31
Unused lines of credit/assets	0.065	0.143*	0.106	0.123
Percent with public debt	21.7		36.0	
Percent with restructured debt***	2.22		5.33	
Number	135		261	

* Significantly different from the multiple bank group at the 0.05 level.
** Calculated only for years 1985 and 1990.
*** Firms that indicated in their financial reports that they restructured their debt in the fiscal year.

(b) Table 4: Firms with one bank vs. multiple banks.

6.4 Table 5: Public debt outstanding

Read:

- Table 5 compares firms with and without public debt.
- Firms with public debt are larger, older, and more highly levered.
- Firms without public debt rely more heavily on bank debt: mean bank debt over total debt is 0.73 versus 0.50 for firms with public debt.
- Public-debt firms are much more likely to have multiple banking relationships: 84 percent versus 54 percent.
- Market-to-book ratios are similar across the two groups, which helps separate access to public debt from growth opportunities.

Economic message: Public debt and multiple bank relationships tend to appear together. This supports the idea that firms with better outside financing options are less exposed to one-bank hold-up, but it also complicates separate identification of the two channels.

6.5 Table 6: Baseline regressions for bank debt reliance

Read:

- Table 6 is the central regression table. The dependent variable is bank loans divided by total debt.
- Market-to-book is negative and statistically significant in specifications without the interaction. This means that, for single-bank firms, growth opportunities are associated with lower reliance on bank debt.
- The interaction between multiple banks and market-to-book is positive and statistically significant.
- R&D-to-sales shows the same qualitative pattern: R&D is negative by itself, but the multiple-bank interaction is positive.
- Log assets and leverage are generally negatively related to bank-debt reliance.
- Year dummies are not economically or statistically central, suggesting no strong secular decline in bank debt after controlling for firm characteristics.

Economic message: The main result is not simply that high-growth firms avoid banks. Rather, high-growth firms avoid being dependent on a single bank. When they have multiple banks, bank debt becomes more compatible with growth opportunities.

Table V

Descriptive Statistics Concerning the Financial Characteristics and Debt Structure for Firms Sorted by whether they have Public Debt Outstanding

The sample consists of observations for 1980, 1985, and 1990 for 250 randomly selected exchange and National Association of Securities Dealers (NASD) traded firms with long term debt outstanding in 1980. Financial information is from COMPUSTAT, 10K filings, and annual reports. Leverage is measured as short plus long term debt divided by total assets. The market to book ratio is the book value of liabilities and preferred stock plus the market value of common stock divided by the book value of assets. Bank debt is from the firm's annual report or 10K filing. Total debt equals short plus long term debt less industrial revenue bonds, mortgages, and capitalized leases. Other private debt is privately placed bonds and loans specified in the financial statements as being from nonbank sources. Investment equals capital expenditures as reported in COMPUSTAT. Fixed assets are plant, property, and equipment.

	Firms Without Public Debt		Firms with Public Debt	
	Mean	Median	Mean	Median
Assets (millions)	528*	63.9*	3127	5
Age	34*	42.1*	47.4	46
Leverage	0.31*	0.33*	0.41	0.51
Coverage ratio	15.38*	3.54	3.94	3.08
R&D/sales	0.04	0.02	0.05	0.02
Market/book	1.42	1.07	1.30	1.08
Bank debt/total debt	0.73*	0.97*	0.50	0.51
Bank debt/assets	0.15	0.11	0.17	0.12
Public debt/total debt	0.00	0.00	0.34	0.40
Commercial paper/total debt	0.00	0.00	0.03	0.00
Other private/total debt	0.14*	0.01	0.07	0.01
Investment/assets	0.07	0.06	0.08	0.07
Fixed assets/assets	0.32*	0.30	0.36	0.30
Growth in Assets over five years**	0.602	0.447	0.955	0.376
Return on assets (ROA)	0.078	0.094	0.081	0.096
Unused lines of credit/assets	0.11	0.063	0.15	0.09
Percent with secured bank debt	30		32	
Percent with multiple banking relationships	54		84	
Percent with restructured debt***	2.8		5.1	
Number	372		213	

* Significantly different from the public debt group at the 0.05 level.
** Calculated only for years 1985 and 1990.
*** Firms that indicated in their financial reports that they restructured their debt in the fiscal year.

(a) Table 5: Firms with and without public debt.

Table VI

Regression Results Relating the Ratio of Book Loans to Total Debt to Firm Characteristics

The regression is based on a sample of 250 publicly traded firms for 1980, 1985, and 1990. The dependent variable is bank loans to total debt. Leverage equals short plus long term debt divided by total assets. Market to book ratio equals book value of assets minus book value of equity plus the market value of equity divided by the book value assets. Multiple banks is a dummy variable that equals one if the firm reports multiple banking relationships in its financial statements. *t*-statistics are in parentheses and for ordinary least squares (OLS) results are based on White corrected standard errors.

	(1)	(2)	(3)*	(4)	(5)	(6)*
Number	358	358	358	158	158	158
Intercept	0.821	0.789	0.915	0.796	0.732	0.769
	(14.202)	(12.967)	(9.612)	(9.802)	(8.293)	(6.192)
Coverage	−0.002	−0.002	−0.002	−0.000	−0.000	−0.000
	(−2.002)	(−1.841)	(−1.722)	(−0.339)	(−0.253)	(−0.092)
Leverage	−0.042	−0.045	−0.057	−0.028	−0.035	−0.032
	(−2.105)	(−2.243)	(−1.711)	(−2.239)	(−2.923)	(−0.699)
Log assets	−0.030	−0.039	−0.063	−0.021	−0.036	−0.050
	(−2.279)	(−3.942)	(−3.770)	(−2.122)	(−3.239)	(−2.460)
Market/book	−0.045	−0.033	−0.051			
	(−3.377)	(−2.503)	(−2.071)			
Multiple banks × Market/book	0.069	0.044	0.066			
	(3.990)	(3.078)	(2.070)			
R&D/sales				−1.967	−1.373	−1.922
				(−3.694)	(−2.182)	(−1.651)
Multiple banks × R&D/sales				2.280	0.800	1.355
				(2.301)	(1.142)	(1.747)
Multiple banks (dummy)		0.126	0.213		0.212	0.278
		(2.259)	(2.553)		(2.339)	(2.333)
Dummy 1985	0.019	0.010	0.020	0.011	0.013	0.051
	(0.453)	(0.241)	(0.271)	(0.154)	(0.182)	(0.517)
Dummy 1990	0.064	0.065	0.070	−0.012	0.017	0.055
	(1.190)	(1.233)	(0.899)	(−0.150)	(0.232)	(0.522)
σ^2 from Tobit estimates			0.517			0.483
			(19.273)			(13.200)
R^2	0.057	0.071		0.062	0.101	

* Double-sided Tobit estimates.

(b) Table 6: Bank-debt regressions and multiple-bank interactions.

6.6 Tables 7 and 8: Robustness and alternative debt structures

Read Table 7:

- Table 7 repeats the main regressions in restricted samples.

- The first two columns require leverage above 0.10.
- The next two columns require bank loans over assets above 0.03.
- The final two columns restrict to firms without public debt.
- The positive multiple-bank interaction with growth opportunities remains in these restricted samples.

Read Table 8:

- Table 8 studies whether public debt or other nonbank private debt moderates the relation between growth opportunities and bank-debt reliance.
- Public debt interacted with market-to-book is positive in the OLS specification, consistent with public debt reducing bank hold-up.
- The public-debt dummy is strongly negative because public debt mechanically substitutes for bank debt in the debt mix.
- Other nonbank private debt does not show the same positive interaction with market-to-book.

Economic message: The main findings survive several sample restrictions. Public debt appears to play some hold-up mitigation role, but other private debt does not look like a close substitute for bank debt or multiple-bank relationships.

Table VII

Regression Results Relating the Ratio of Bank Loans to Total Debt to Firm Characteristics for Firms Meeting Minimum Leverage and Bank Loan to Asset Ratios and for Firms without Public Debt

The regression is based on a sample of 250 publicly traded firms for 1980, 1985, and 1990. The dependent variable is bank loans to total debt. Leverage equals short plus long term debt divided by total assets. Market to book ratio equals book value of assets minus book value of equity plus the market value of equity divided by the book value assets. Multiple banks is a dummy variable that equals one if the firm reports multiple banking relationships in its financial statements. *t*-statistics are in parentheses and are based on White corrected standard errors.

	Firms with Leverage Greater Than 0.10		Firms with Bank Loans/Assets Greater Than 0.03		Firms with No Public Debt	
	(1)	(2)	(3)	(4)	(5)	(6)
Number	319	125	278	124	203	87
Intercept	0.809	0.811	1.045	1.057	0.787	0.722
	(9.294)	(6.421)	(23.92)	(14.488)	(9.387)	(6.191)
Coverage	0.000	0.000	0.0046	0.0023	−0.001	0.001
	(0.214)	(0.097)	(2.762)	(1.139)	(−0.593)	(0.702)
Leverage	−0.089	−0.052	−0.037	−0.030	−0.022	−0.014
	(−0.845)	(0.645)	(−1.423)	(−1.493)	(−1.616)	(−1.014)
Log assets	−0.040	−0.032	−0.055	−0.044	−0.009	−0.013
	(−3.783)	(−2.431)	(−7.113)	(−4.220)	(−0.485)	(−0.595)
Market/book	−0.027		−0.021		−0.031	
	(−1.881)		(−3.103)		(−2.318)	
Multiple banks × Market/book	0.034		0.020		0.046	
	(2.234)		(2.183)		(2.509)	
Multiple banks (dummy)	0.123	0.124	0.019	0.000	0.099	0.140
	(2.126)	(1.149)	(0.463)	(0.001)	(1.540)	(1.414)
R&D/sales		−1.355		−1.418		−1.211
		(−1.858)		(−1.045)		(−1.572)
Multiple banks × R&D/sales		0.764		1.809		2.298
		(1.612)		(1.573)		(2.189)
Dummy 1985	0.015	−0.025	−0.014	−0.066	0.070	0.095
	(0.363)	(0.313)	(−0.473)	(−1.312)	(1.427)	(1.087)
Dummy 1990	0.054	−0.025	−0.071	−0.082	0.054	0.084
	(0.984)	(−0.339)	(1.842)	(−1.366)	(0.886)	(0.892)
R^2	0.06	0.07	0.210	0.216	0.095	0.154

(a) Table 7: Restricted-sample bank-debt regressions.

Table VIII

Regression Estimates Relating the Ratio of Bank Loans to Total Debt to Firm Financial Characteristics and Whether the Firm Uses Public Debt or Nonbank Private Debt

The regression is based on a sample of 250 publicly traded firms for 1980, 1985, and 1990. The dependent variable is bank loans to total debt. Leverage equals short plus long term debt divided by total assets. Market to book ratio equals book value of assets minus book value of equity plus the market value of equity divided by the book value of assets. Public debt equals one if the firm has public debt outstanding. Other private equals one if the firm has nonbank private debt outstanding. *t*-statistics are in parentheses and for ordinary least squares (OLS) results are based on White corrected standard errors.

	(1)	(2)*	(3)	(4)*
Number	502	502	531	531
Intercept	0.746	0.840	0.825	0.795
	(16.374)	(12.419)	(18.263)	(16.773)
Coverage	−0.000	0.001	−0.001	0.001
	(−0.064)	(0.611)	(−1.743)	(1.340)
Leverage	−0.014	−0.003	−0.018	−0.044
	(−0.777)	(−0.073)	(−0.810)	(−1.521)
Log assets	0.022	0.325	−0.009	−0.015
	(2.860)	(2.788)	(−1.260)	(−1.947)
Market/book	−0.017	−0.028	−0.003	−0.017
	(−1.597)	(−1.675)	(−0.345)	(−1.540)
Public debt (dummy) × market/book	0.027	0.039		
	(2.338)	(1.571)		
Public debt (dummy)	−0.387	−0.578		
	(−11.025)	(−9.947)		
Other private debt (dummy) × market/book			−0.008	−0.060
			(−0.538)	(−0.634)
Other private debt (dummy)			−0.017	0.022
			(−0.222)	(0.183)
Dummy 1985	0.020	0.040	0.010	0.054
	(0.661)	(0.857)	(0.283)	(0.990)
Dummy 1990	−0.017	−0.043	−0.044	−0.720
	(−0.434)	(−0.813)	(−1.006)	(−0.176)
σ^2 from Tobit estimates		0.431		0.488
		(25.532)		(23.250)
R^2	0.212		0.085	

(b) Table 8: Public debt, other private debt, and bank-debt reliance.

7 Short summary

- The paper studies how large publicly traded firms mix bank debt, public debt, commercial paper, and other private debt.
- The authors hand-collect debt-structure information from annual reports and 10-K filings for 250 firms in 1980, 1985, and 1990.
- Bank debt is the largest source of debt financing in the sample, with an overall mean bank-debt-to-total-debt ratio of about 0.64.
- The key theoretical tension is that banks monitor and renegotiate efficiently, but a single informed bank can create an information monopoly.
- For firms with a single bank relationship, reliance on bank debt is negatively related to market-to-book and R&D intensity.
- For firms with multiple bank relationships, the relation between growth opportunities and bank debt is much less negative and often positive.
- This pattern is consistent with the idea that multiple banks reduce hold-up problems.
- Public debt also appears to mitigate bank hold-up, although it is difficult to separate from multiple banking because many public-debt firms also have multiple banks.
- Other nonbank private debt does not appear to perform the same role.
- The paper's core message is that relationship lending is valuable but can become costly when the borrower is locked into one bank.

8 Conclusion

8.1 Core conclusion

Houston and James show that bank information monopolies matter for corporate debt structure. Firms with valuable growth opportunities do not rely heavily on a single bank lender. They use multiple banking relationships, public debt, or other financing arrangements to limit bank bargaining power.

8.2 Economic intuition

Banks are valuable because they monitor borrowers and can renegotiate flexibly. But these same strengths give banks private information and control rights. When only one bank has this information, the firm can be held up in future financing. Growth firms are especially vulnerable because future projects create surplus that the bank can try to capture.

8.3 Incremental contribution revisited

The paper moves the bank-lending literature beyond the idea that bank loans are simply special. It shows that the structure of bank relationships matters. Bank debt can be attractive when lender competition or public-market access limits hold-up, but unattractive when a single bank has monopoly power over information.

8.4 Implications

The results help explain why large firms often maintain a mixture of private and public debt. A mixed debt structure can preserve the monitoring and flexibility benefits of bank debt while reducing the cost of being dependent on one informed lender.

8.5 Limitations

The data are hand-collected and rich, but the design is observational. Multiple-bank firms differ systematically from single-bank firms; public-debt firms differ from firms without public debt. Bank debt is also measured broadly, so some unidentified nonbank private debt may be counted as bank debt. These issues make the results best interpreted as strong evidence consistent with bank hold-up rather than a fully causal estimate of one bank's bargaining power.

8.6 Takeaway

The enduring takeaway is that the value of bank debt depends on the market structure of lending relationships. Banks solve information problems, but they can also create them. Firms with growth opportunities protect themselves by avoiding dependence on a single informed bank.